

Multiple comparisons

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Statistical tests and assumptions

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Uncertainty intervals

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Experimental units and independence

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Multiple comparisons. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Multiple comparisons. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Four tests have p-values .01, .02, .2, .9. Compute threshold and decisions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Five tests have p-values .009, .012, .3, .4, .8. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Four tests have p-values .01, .02, .2, .9. Compute threshold and decisions.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled multiple comparisons evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of multiple comparisons. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Multiple comparisons

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

Familywise $\alpha=0.05$. Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

Supplied p-values and six-test decisions

	Supplied p	At cutoff 0.00833
Test 1	0.004	Reject
Test 2	0.012	Do not reject
Test 3	0.02	Do not reject
Test 4	0.08	Do not reject
Test 5	0.4	Do not reject
Test 6	0.9	Do not reject

Read a cell using both its row and column labels.

Multiple comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count the planned tests

There are six tests in the declared family.



2. Compute corrected threshold

$0.05/6$ is about 0.00833 for each test.



3. Apply all six decisions

Only 0.004 is below 0.00833; the other five do not reject under this rule.

CHANGE ONE THING

The declared family has ten tests instead of six; keep the listed values unchanged.

Explain the changed result using the matching lesson.

Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only 0.004 is below 0.00833; the other five do not reject under this rule.

1. Apply all six decisions
2. Compute corrected threshold
3. Count the planned tests

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply all six decisions Only 0.004 is below 0.00833; the other five do not reject under this rule. Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required. A rejected null is not automatically a large or useful improvement.

Practice 2: explain the deciding evidence

Changed-case practice: The declared family has ten tests instead of six; keep the listed values unchanged. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects. The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

Repair a missing building block

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the mechanism and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum. Count null arrangements with absolute sum at least four, compute p , and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the mechanism and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Experimental units and independence (ai-7/9). Goal: Explain the mechanism and terminology of experimental units and independence. Short practice: Count independent plants and photo views. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
